

Table 1: GP-FL Performance by Feature and Cluster with SVGP for Dataset 1

Feature	Cluster	Model	Accuracy	Precision	Recall	F1-Score
gender	Female	GP-FL	0.9368	1.0000	0.1435	0.2510
	Male	GP-FL	0.9211	0.9697	0.1978	0.3285
age	Younger	GP-FL	0.9825	0.0000	0.0000	0.0000
	Older	GP-FL	0.8719	1.0000	0.1387	0.2436
hypertension	No Hypertension	GP-FL	0.9393	1.0000	0.1022	0.1855
	Hypertension	GP-FL	0.8691	0.9074	0.5889	0.7143
heart_disease	No Heart Disease	GP-FL	0.9273	1.0000	0.0238	0.0464
	Heart Disease	GP-FL	0.8492	0.8483	0.6214	0.7173
smoking_history	Never	GP-FL	0.9548	0.9851	0.5140	0.6755
	Former	GP-FL	0.9071	0.9569	0.3578	0.5208
	Current	GP-FL	0.9376	0.9747	0.4031	0.5704
	No Info	GP-FL	0.9707	1.0000	0.2364	0.3824
bmi	Underweight	GP-FL	0.9923	0.0000	0.0000	0.0000
	Healthy	GP-FL	0.9624	0.0000	0.0000	0.0000
	Overweight	GP-FL	0.9291	1.0000	0.0122	0.0241
	Obesity I	GP-FL	0.8717	0.9556	0.1168	0.2082
	Obesity II	GP-FL	0.8479	0.9296	0.2651	0.4125
	Obesity III	GP-FL	0.8692	0.8623	0.5385	0.6630
HbA1c_level	Normal	GP-FL	1.0000	0.0000	0.0000	0.0000
	Pre-diabetes	GP-FL	0.9180	1.0000	0.0029	0.0059
	Diabetes	GP-FL	0.9200	0.9641	0.6854	0.8012
blood_glucose_level	Normal	GP-FL	1.0000	0.0000	0.0000	0.0000
	Pre-diabetes	GP-FL	1.0000	0.0000	0.0000	0.0000
	Diabetes	GP-FL	0.8976	1.0000	0.1133	0.2036

Table 2: Global Model Summary (All Features) with SVGP for Dataset 1

Feature	Accuracy	Precision	Recall	F1-Score
gender	0.9303	0.9827	0.1698	0.2895
age	0.9268	1.0000	0.1243	0.2212
hypertension	0.9340	0.9372	0.2236	0.3611
heart_disease	0.9243	0.8726	0.1106	0.1963
smoking_history	0.9495	0.9767	0.4031	0.5707
bmi	0.9268	0.9008	0.1409	0.2437
HbA1c_level	0.9495	0.9642	0.4057	0.5711
blood_glucose_level	0.9265	1.0000	0.1133	0.2036

Table 3: GP-FL Performance by Feature and Cluster with SVGP for Dataset 2

Feature	Cluster	Model	Accuracy	Precision	Recall	F1-Score
Pregnancies	Cluster 0 (0.0-4.0)	GP-FL	0.8182	0.7222	0.5000	0.5909
	Cluster 1 (5.0-17.0)	GP-FL	0.6786	0.6571	0.7931	0.7188
Glucose	Normal	GP-FL	0.8919	0.0000	0.0000	0.0000
	Pre-diabetes	GP-FL	0.7455	0.4167	0.4167	0.4167
	Diabetes	GP-FL	0.6667	0.8077	0.5833	0.6774
BloodPressure	No Info	GP-FL	0.7143	0.6667	0.6667	0.6667
	Low	GP-FL	0.8208	0.7586	0.6471	0.6984
	Normal	GP-FL	0.6207	0.6667	0.4286	0.5217
	Elevated	GP-FL	0.6667	0.6000	1.0000	0.7500
SkinThickness	Cluster 0 (37.0-99.0)	GP-FL	0.7407	0.6875	0.8462	0.7586
	Cluster 1 (0.0-8.0)	GP-FL	0.7447	0.7500	0.5000	0.6000
	Cluster 2 (10.0-24.0)	GP-FL	0.8333	0.6000	0.7500	0.6667
	Cluster 3 (25.0-36.0)	GP-FL	0.7609	0.6875	0.6471	0.6667
Insulin	No Info	GP-FL	0.7200	0.7647	0.4333	0.5532
	Normal	GP-FL	0.8077	0.6500	0.8125	0.7222
	High	GP-FL	0.7407	0.6000	0.9000	0.7200
BMI	Healthy	GP-FL	0.9048	0.6667	0.6667	0.6667
	Overweight	GP-FL	0.7778	0.6000	0.6000	0.6000
	Obesity I	GP-FL	0.7333	0.6190	0.7647	0.6842
	Obesity II	GP-FL	0.8000	0.7500	0.7500	0.7500
	Obesity III	GP-FL	0.7500	0.7059	1.0000	0.8276
DiabetesPedigreeFunction	Cluster 0 (0.6-2.4)	GP-FL	0.7179	0.6800	0.8500	0.7556
	Cluster 1 (0.1-0.6)	GP-FL	0.8000	0.7778	0.5526	0.6462
Age	Younger	GP-FL	0.8784	0.6250	0.7692	0.6897
	Older	GP-FL	0.6667	0.6731	0.7778	0.7216

Table 4: Global Model Summary (All Features) with SVGP for Dataset 2

Feature	Accuracy	Precision	Recall	F1-Score
Pregnancies	0.7419	0.6230	0.6909	0.6552
Glucose	0.7171	0.5634	0.7692	0.6504
BloodPressure	0.7403	0.6349	0.7018	0.6667
SkinThickness	0.7500	0.6349	0.7143	0.6723
Insulin	0.6818	0.5660	0.5357	0.5505
BMI	0.7632	0.7045	0.5741	0.6327
DiabetesPedigreeFunction	0.7597	0.6721	0.7069	0.6891
Age	0.7290	0.6290	0.6724	0.6500

Table 5: GP-FL Performance by Feature and Cluster with SVGP for Dataset 3

Feature	Cluster	Model	Accuracy	Precision	Recall	F1-Score
HighBP	No High BP	GP-FL	0.9126	0.0000	0.0000	0.0000
	High BP	GP-FL	0.7122	0.0000	0.0000	0.0000
HighChol	Normal Cholesterol	GP-FL	0.8911	0.0000	0.0000	0.0000
	High Cholesterol	GP-FL	0.7336	0.0000	0.0000	0.0000
BMI	Underweight	GP-FL	0.9378	0.0000	0.0000	0.0000
	Healthy	GP-FL	0.9179	0.0000	0.0000	0.0000
	Overweight	GP-FL	0.8474	0.0000	0.0000	0.0000
	Obesity I	GP-FL	0.7692	0.0000	0.0000	0.0000
	Obesity II	GP-FL	0.6882	0.3529	0.0047	0.0093
	Obesity III	GP-FL	0.6323	0.6204	0.0647	0.1172
HeartDiseaseorAttack	No Heart Disease	GP-FL	0.8430	0.0000	0.0000	0.0000
	Heart Disease	GP-FL	0.6404	0.6105	0.0336	0.0637
GenHlth	Excellent	GP-FL	0.9565	0.0000	0.0000	0.0000
	Very Good	GP-FL	0.9038	0.0000	0.0000	0.0000
	Good	GP-FL	0.7991	0.0000	0.0000	0.0000
	Fair	GP-FL	0.6556	0.5806	0.0247	0.0473
	Poor	GP-FL	0.6184	0.6522	0.0934	0.1633
PhysHlth	Good (0-13 days)	GP-FL	0.8450	0.0000	0.0000	0.0000
	Poor (14+ days)	GP-FL	0.6950	0.5714	0.0019	0.0039
DiffWalk	No Difficulty Walking	GP-FL	0.8608	0.0000	0.0000	0.0000
	Difficulty Walking	GP-FL	0.6592	0.6344	0.0201	0.0390
Age	Younger	GP-FL	0.9046	0.0000	0.0000	0.0000
	Older	GP-FL	0.7643	0.0000	0.0000	0.0000

Table 6: Global Model Summary (All Features) with SVGP for Dataset 3

Feature	Accuracy	Precision	Recall	F1-Score
HighBP	0.8216	0.0000	0.0000	0.0000
HighChol	0.8215	0.0000	0.0000	0.0000
BMI	0.8224	0.5840	0.0089	0.0176
HeartDiseaseorAttack	0.8221	0.6105	0.0071	0.0140
GenHlth	0.8291	0.6234	0.0182	0.0354
PhysHlth	0.8230	0.5714	0.0005	0.0010
DiffWalk	0.8234	0.6344	0.0072	0.0143
Age	0.8188	0.0000	0.0000	0.0000